

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

SOCIAL NETWORK ANALYSIS

(CSE (Data Science))

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

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| (a) What is Lexical Diversity? How is it measured? | 2M |
| (b) What is XFN? List any two applications of XFN? | 2M |
| (c) What is Mail Analysis? | 2M |
| (d) What is the logic behind finding the similarity between two people on Twitter? | 2M |
| (e) List some of the Statistics you would like to collect when you are analyzing your favourite actors tweets. | 2M |
| (f) Does a frequently appearing user entity imply friendship? | 2M |
| (g) Explain any two-similarity metrics for clustering LinkedIn. | 2M |
| (h) What is Dorling Cartogram? How does it help to map the professional network? | 2M |
| (i) What is R Graph? | 2M |
| (j) Explain the Python SDK officially maintained by Face book for Graph API. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|-----------|--|----|
| 2 | (a) During the election season, a party wants to know what people are talking about their party and opposition party on twitter. Can you give the code snippet to help them do it? | 5M |
| | (b) How can we use Breadth First Crawl of XFN data for basic web scrapping application? | 5M |
| OR | | |
| 3 | (a) Along with code snippet, explain tinkering with Twitter's API. | 5M |
| | (b) How, plotting geo data through micro format can help us in making web pages more interesting. | 5M |
| 4 | (a) Explain along with code snippet, the steps to perform Bulk Loading of the documents into CouchDB. | 5M |
| | (b) What is OAuth? How can it be used to authenticate and grab the friend's data? | 5M |
| OR | | |
| 5 | (a) For a sample term of interest, show the usage of SMILE timeline and its output. | 5M |
| | (b) What is a Friendship Clique? How do you Detect and Analyses it? | 5M |
| 6 | (a) Demonstrate a python code snippet to count the number of times a message has been retweeted. | 5M |
| | (b) Write a note on "rereduce" parameter. | 5M |
| OR | | |
| 7 | (a) Discuss the variables used to measure how influential you are as a tweeter. | 5M |
| | (b) Analyses the ways to compare latent social network. | 5M |
| 8 | (a) Explain the motivating factors for clustering your professional network in LinkedIn. | 5M |
| | (b) Differentiate Hierarchical and k-means clustering approaches. | 5M |

OR

Contd. in Page 2

- 9 (a) Demonstrate a python code snippet for Geocoding the locations of your LinkedIn contacts. 5M
(b) Analyses the Greedy Heuristic for clustering the professional network. 5M
- 10 (a) Discuss the Graph API for exploring Face book. 5M
(b) Demonstrate the usage of Tag Cloud to visualize the Wall Data. 5M
- OR**
- 11 (a) Explain the steps to visualize the Face book data using Sunburst. 5M
(b) Demonstrate the usage of FQL query for extracting data about the friends of the logged in user. 5M
